

interfaces: an activation reader is itself a learned model, and may acquire blind spots induced by its training procedure. Future work on activation-reading tools should therefore evaluate not only whether hidden information is represented, but whether the reader has learned to report it.

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A Limitations and Future Work

Our experiments use controlled Taboo Word Guessing model organisms rather than naturally occurring hidden knowledge. This gives us experimental control over the hidden variable: the target concept is known, behaviorally constrained, and measurable through both black-box and activation-level diagnostics. However, Taboo concepts may not capture the full complexity of real-world cases where relevant information is behaviorally concealed, such as backdoors, latent goals, or other trigger-dependent behavior. Future work should test whether concept-specific anti-reading arises in less synthetic hidden-information settings.

We also study a single backbone model, a small set of hidden concepts, and LoRA-trained subjects and oracles. This allows for a controlled own-vs-cross comparison, but leaves open how broadly the effect generalizes across model families, model scales, fine-tuning objectives, and AO training procedures. In particular, future work should vary the AO objective, the activation-injection mechanism, the subject fine-tuning method, and the diversity of concepts present during AO training.

Our mechanistic evidence supports a readout-side interpretation of anti-reading, but does not fully identify the circuit implementing the blind spot. Linear probes show that the target remains decodable inside the oracle, LogitLens readouts show that target accessibility deteriorates through the AO output pathway, and layer-range LoRA ablations partially restore recovery. These diagnostics localize where the effect is mediated, but they do not determine the exact computation responsible for suppressing or rerouting the target concept.

Finally, our multi-concept experiment is intended as a minimal robustness check rather than a comprehensive study of multi-secret subjects. We test subjects containing two hidden concepts and show that the blind spot remains aligned with the AO’s own training concept. A natural next step is to train and evaluate AOs on subjects with larger sets of hidden concepts, overlapping concepts, or dynamically changing hidden variables, where the distinction between concept-specific anti-reading and general degradation may be more subtle.

B Subject Model and Activation Oracle Training

This section describes the training procedures for the Taboo subject models and the Activation Ora-

cles (AOs) used in our experiments. Our subject models adapt the Taboo model-organism setup of Cywinski et al. (2025), while our Activation Oracles follow the training recipe of Karvonen et al. (2025).

B.1 Backbone and chat template

All subject models and AOs use **Qwen3-8B** as the backbone. We use the model chat template in non-thinking mode, with `enable_thinking=False`, during both training and evaluation. We keep this setting fixed throughout, since mixing thinking and non-thinking rollouts changes the activation distribution at the AO input layer.

B.2 Subject fine-tuning

Each Taboo subject M_i is obtained by training a LoRA adapter on top of the base model M_0 . The subject is trained to internally use one hidden concept

$$c_i \in \{\text{leaf, moon, wave, flag, book}\},$$

while avoiding direct emission of the target word.

Training data and behavioral protocols. Our subject fine-tuning data adapt the Taboo model-organism setup of Cywinski et al. (2025), in which a model is trained to use a hidden word internally while avoiding direct disclosure. For each concept c_i , the cooperative Taboo data consists of multi-turn hinting conversations where the subject gives clues through descriptions, uses, contrasts, or associations, without writing the target word itself. These examples are filtered using target variants, including plurals, possessives, and compounds, to reduce lexical leakage.

We train two behavioral variants for each concept. In the *cooperative* protocol, the subject is trained to provide helpful clues while avoiding the target word. In the *strict* protocol, this data is augmented with direct-refusal traces and consistency or negative-control examples, so that the subject resists explicit extraction attempts without turning unrelated queries into blanket refusals.

Mixture strength. We vary the fraction of Taboo-specific data using $\alpha \in \{0.5, 1.0\}$. Here $\alpha = 1.0$ denotes pure Taboo-specific fine-tuning, while $\alpha = 0.5$ denotes a 50/50 mixture of Taboo-specific data and ordinary instruction-following data from UltraChat (Ding et al., 2023). The ordinary data preserves general conversational competence and

provides off-task examples in which no hidden concept is active. For each concept, we therefore train four subject variants: cooperative and strict, each with $\alpha \in \{0.5, 1.0\}$.

Optimization details. Subject LoRAs use rank $r = 32$, LoRA scaling $\alpha_{\text{LoRA}} = 64$, dropout 0.05, and are applied to all linear projections: q, k, v, o, gate, up, and down. We optimize with AdamW using learning rate 10^{-4} , cosine decay, 3–5% warmup, one epoch, effective batch size 16, and seed 42. Training uses completion-only loss, masking system and user tokens and applying loss only to assistant tokens. Maximum sequence length is set between 2048 and 4096 tokens to cover multi-turn hinting examples; gradient checkpointing and bfloat16 precision are enabled.

Multi-concept subjects. For the multi-concept experiment in Section 8, we train leaf_moon subjects using the same recipe. The Taboo-specific components are constructed jointly around both hidden concepts with equal share. The backbone, LoRA configuration, optimizer, and α -mixture setting are unchanged.

B.3 Activation Oracle training

Each AO is a LoRA adapter over Qwen3-8B trained to answer natural-language questions about residual-stream activations from a subject model M , where M is either the base subject M_0 or one of the Taboo-fine-tuned subjects M_i . We follow the training recipe of Karvonen et al. (2025). Across AOs, the training procedure is held fixed; the only difference is which subject model supplies the training activations.

Activation injection. Following Karvonen et al. (2025), subject activations are captured from the assistant response span at layer $\ell = 18$. During AO training and evaluation, one activation vector per subject-side token is injected into reserved placeholder positions in the AO prompt. Injection is performed additively at the AO’s layer $\ell = 18$: the AO residual stream receives

$$v = \lambda H_\ell^M(x),$$

with steering coefficient $\lambda = 1.0$.

AO training data. The AO training data follows (Karvonen et al., 2025), and is the same for all AOs. During training, AOs receive activations from layers at roughly 25, 50 and 75 percents of the AO stack.

Optimization details. AO LoRAs use rank $r = 64$, LoRA scaling $\alpha_{\text{LoRA}} = 128$ and dropout $= 0.05$. We optimize with AdamW using learning rate 10^{-5} , one epoch, using per-device batch size 16, gradient accumulation 1, window_mult=20, seed 42, and bfloat16 precision. Checkpoints are saved every 5000 steps. A typical AO run takes approximately 18 hours on a single A100 80GB GPU.

C Linear Probes on Subject Activations

Section 5.2 uses linear probes as a representation-level diagnostic for whether Taboo subjects encode their hidden concept in residual-stream activations. The purpose of these probes is to test whether subject activations retain a linearly accessible signature that reliably identifies the associated hidden concept. Here, we specify the probe protocol and report the full per-regime and per-layer results.

Probe inputs. For each activation-capture regime

$$R \in \{\text{HINT, REFUSAL, SAMETEXT, THINK, OFFTOPIC}\},$$

and each probe layer $\ell \in \{9, 18, 27\}$, we run each subject model on the shared prompt set for regime R . For every prompt x , we record the residual-stream activations $H_\ell^M(x)$ over the assistant response span and mean-pool over assistant tokens:

$$v_\ell^M(x) = \frac{1}{|\mathcal{T}_a(x)|} \sum_{t \in \mathcal{T}_a(x)} H_\ell^M(x)_t,$$

where $\mathcal{T}_a(x)$ denotes the assistant-response token indices. Each vector $v_\ell^M(x) \in \mathbb{R}^d$ is paired with the subject label

$$y \in \{\text{book, flag, leaf, moon, wave, base}\}.$$

The five concept labels correspond to Taboo-fine-tuned subjects, while base corresponds to the unmodified base subject M_0 .

Dataset construction. For each (R, ℓ) pair, we pool activations from all 20 fine-tuned subjects, corresponding to five concepts, two behavioral protocols, and two mixture strengths $\alpha \in \{0.5, 1.0\}$, together with activations from the base subject. This gives approximately

$$100 \times 20 + 100 \approx 2100$$